

Petalino Biologino: The Fold Code

花瓣仔：折疊密碼

1



A mysterious box!
神秘寶箱！

2



Wrong shape...
won't open!
形狀唔啱.....
打唔開！

3



The lock needs
the right protein
shape.
鎖要啱形蛋白質
先開得。

SHAPE
MATTERS
形狀好重要！



4



Proteins come from
amino acids.
蛋白質來自胺基酸。

Teach me
the Fold Code!
教我折疊密碼！

MISSION: Code the right protein & unlock the box
任務：編碼啱形蛋白質，打開寶箱

1

AMINO ACIDS = BUILDING BLOCKS

胺基酸 = 積木

AMINO FORGE
胺基酸鍛造坊

BUILD LIFE!
創造生命!

First: building
blocks for the
lock code.

第一步：
砌鎖碼嘅積木。

2

AMINO ACIDS

胺基酸

Building blocks
of proteins!

蛋白質嘅基本單位!



3

20 TYPES

20種

Different side
chains (R)
唔同側鏈 (R)



4

Wrong blocks →
wrong shape →
lock stays shut.

積木錯 →
形狀錯 →
鎖開唔到。

Next, link
them!
下一步：
連接!



AMINO ACIDS = BUILDING BLOCKS · 20 TYPES

胺基酸 = 積木 · 20種

1

DIPEPTIDE GATE / 二肽關

CONDENSATION / 縮合

Start coding
the password chain.
開始編碼
密碼鏈。



2

CONDENSATION = form bonds + release water
縮合 = 成鍵 + 釋出水



3

peptide bond / 肽鍵



+

1 H₂O / 1水

2 amino acids → dipeptide + 1 H₂O
2 胺基酸 → 二肽 + 1 水

4

First
password
segment!
第一段
密碼!

2 AA →
1 DIPEPTIDE
+ 1 WATER



POLYPEPTIDE FACTORY / 多肽廠

PASSWORD STRING / 密碼繩



1

Many amino acids →
polypeptide password string!
好多胺基酸 → 多肽密碼繩!

POLYPEPTIDE STRING
多肽密碼繩

AMINO
ACIDS
胺基酸



LINK!
連接!

z
z
z

2

Bond Math / 鍵結計算

$n \text{ AA} \rightarrow (n-1) \text{ bonds}$

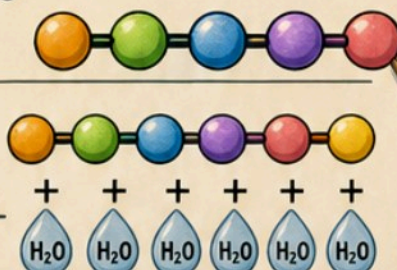
$n \text{ 胺基酸} \rightarrow (n-1) \text{ 鍵}$

$5 \text{ AA} \rightarrow 4 \text{ bonds}$

$5 \text{ 胺基酸} \rightarrow 4 \text{ 鍵}$

Break to $6 \text{ AA} \rightarrow 5 \text{ H}_2\text{O}$

水解成 6 胺基酸 → 5 個水分子



Linear chain = inactive / still can't unlock.

線性長鏈 = 無活性 / 仲開唔到鎖。

z
z
z

Fold Dial

STRING
繩狀

4

Chain isn't enough —
fold the right 3D shape.

有鏈未夠，
要摺成啱嘅立體形。

PROTEIN LOCK
蛋白質鎖

CHAIN FIRST • FOLD TO UNLOCK

先有鏈，折疊先開鎖

FOLD CHAMBER / 折疊室

RIGHT SHAPE UNLOCKS / 啱形開鎖

1

Folding / 折疊

Polypeptide
多肽鏈

Hydrogen
bond clips
氫鍵夾

Folding makes
the right shape!
折疊造就
正確形狀!

2

Polypeptide → Protein (3D) / 多肽 → 蛋白質 (立體)

Sequence
determines shape!
序列決定構象!

Fold Dial
折疊轉盤

FOLDED
已折疊

Protein (3D)
蛋白質 (立體)

3

Proteins have many functions!
蛋白質功能多多!

Right shape = correct key!
啱形狀 = 正確的鑰匙!

Perfect fit!
完全吻合!



Growth & Repair
生長修復



Enzymes
酶



Hormones
荷爾蒙



Proto
(Protein Key)
Proto
(蛋白質鑰匙)



PROTEIN LOCK
蛋白質鎖

4

CLICK!

OPEN!
打開!

It opened!
Shape is the code!
開咗!
形狀就係密碼!

Right shape
unlocks.
啱形先開到鎖。

3D SHAPE → UNLOCK
立體形狀 → 開鎖

GUARD THE KEY / 守護鑰匙

DENATURE vs HYDROLYSIS / 變性 vs 水解

TOO HOT!
太熱啦!TOO ACIDIC!
太酸啦!pH
2PROTO PROTEIN KEY
原型蛋白鑰匙IT UNRAVELS!
解開晒!HIGH TEMP
高溫EXTREME pH
極端 pHFold Dial
RUINED
壞咗!MYSTERIOUS
BOX
神秘寶箱**DENATURE** = lose 3D shape → lose unlock function

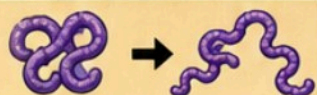
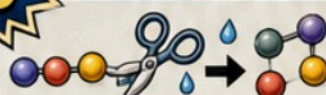
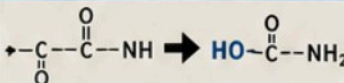
變性 = 失去形狀 → 開唔到鎖

CAUSES: High Temperature · Non-optimal pH
成因: 高溫 · 極端或非最佳 pH**IRREVERSIBLE**
不可逆轉

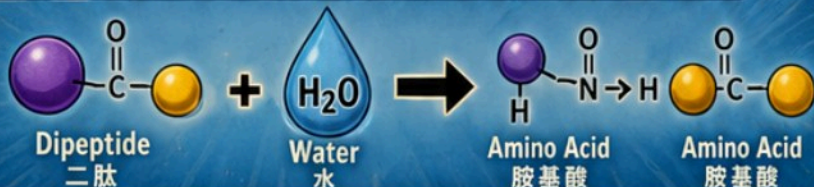
DENATURATION / 變性



HYDROLYSIS / 水解

Affects **SHAPE** (3D structure)
影響形狀 (立體結構)Does **NOT** need water
唔需要水Does **NOT** break peptide bonds
唔會斷開肽鍵**NO** amino acids released
唔會釋放胺基酸Usually **IRREVERSIBLE**
通常不可逆轉Affects **BONDS** (primary structure)
影響鍵結 (初級結構)**NEEDS** water
需要水**BREAKS** peptide bonds
斷開肽鍵**AMINO ACIDS** released
釋放胺基酸Can be **REVERSED** (in theory)
可以 (理論上) 逆轉

HOW HYDROLYSIS WORKS / 水解點運作

Dipeptide
二肽Water
水Amino Acid
胺基酸Amino Acid
胺基酸Dipeptide
二肽

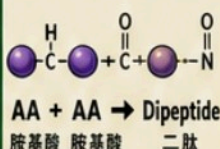
+

H₂O
水

→

2 Amino Acids
兩個胺基酸**BREAKS THE CHAIN, BUT
CAN RESET & REFOLD!**拆開條鏈，
可以重置再折疊!

1 BUILD / 建造

Condensation links
amino acids.
縮合連接胺基酸。AA + AA → Dipeptide
胺基酸 胺基酸 二肽

2 FOLD / 折疊

Right shape unlocks
the box!
正確形狀打開寶箱!

3 WARN / 警告

Heat or pH can **DENATURE**.
Water can **HYDROLYZE**.
高溫或 pH 會令變性，
水解會拆鏈重置。

Link to build · Fold to unlock · Heat/pH ruin shape · Water breaks chain to reset
縮合連接 · 折疊開鎖 · 高溫/極端 pH 破壞形狀 · 水解拆鏈重置

Dr. Loop
迴圈博士